

SYSTEM SPECIFICATION

TicketWave Architecture

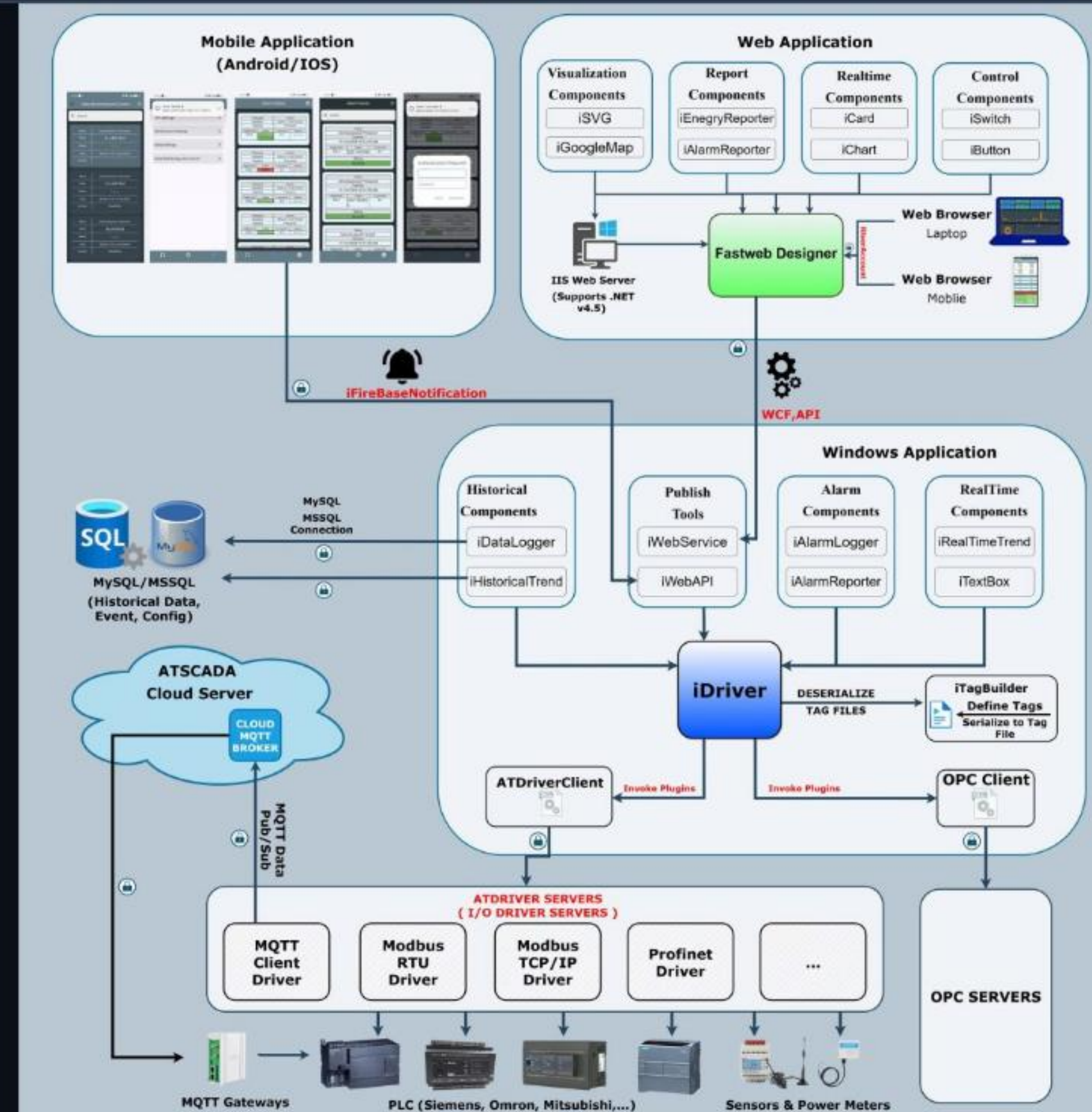
An end-to-end overview of runtime topology, backend layering rules, security models, and cross-cutting system constraints.

01. Runtime Topology

Stateless backend, single-tier SPA, and strict operational constraints

Runtime & Infrastructure

- **Frontend:** Angular 22 SPA served on `:4201` (proxied via `proxy.conf.json` in dev).
- **Backend:** Spring Boot 4.0.6 stateless API running on `:8081` with JWT-bearer security.
- **Database:** PostgreSQL primary with optional read-replica connection routing.
- **In-Process Stack:** Embedded Caffeine cache (30s TTL) and token-bucket rate limiter.



Operational Constraints

Zero External Brokers

Designed with intentional architectural bounds: no Redis, no external message queues, and no live payment gateway. Card decline logic is handled in-process via `CardDeclineSimulator` for predictable test environments.

In-Process Management

Caching, dynamic rate limiting, and seat hold expirations run entirely within the JVM process memory. This guarantees ultra-fast local evaluation while maintaining minimal infrastructure overhead.

02. Backend Layering

Strict one-way dependency flow, package modularity, and explicit domain rules

Backend Layering Rules



Controller

Handles HTTP routing, status codes, OpenAPI metadata, and `@Valid` request binding. Business logic and direct entity access are strictly prohibited.



Service

Contains all business logic, transactional boundaries (`@Transactional`), and security checks (`@PreAuthorize`). MapStruct converts DTOs to Entities.



Repository & Entity

Spring Data JPA queries and specifications. Entities represent the persistence boundary and are never exposed directly to controllers or public APIs.

Feature Package Inventory

Package	Domain Ownership	Key Components
<code>auth</code> & <code>user</code>	Identity & Account Management	JWT filtering, BCrypt hashing, profiles, passenger cards
<code>catalog</code> & <code>booking</code>	Inventory & Reservations	Routes, schedules, seat holds, booking lifecycle, PNR generator
<code>payment</code> & <code>pricing</code>	Financial Transactions	Fare rules, 3DS simulation, promo codes, cancellation refund policy
<code>partner</code> & <code>ledger</code>	Integration & Accounting	OAuth2 M2M credentials, webhooks, append-only financial ledger

03. Security & Governance

Stateless JWT authentication, fine-grained RBAC, rate limiting, and exception safety

Security & Access Control



Stateless JWT

HS256 signature verification requiring `JWT_SECRET` (≥ 32 chars). Fails startup fast if secret is missing or under length per RFC 7518.



Method Security

`@PreAuthorize` role checks for 4 human roles (CUSTOMER, OPERATOR, SUPPORT, ADMIN) and non-human PARTNER_API. Includes tenant isolation checks.



Rate Limiting Filter

Runs prior to Spring Security filter chain. Uses token-bucket limits based on client IP for public routes or `clientId` for partner M2M requests.

Persistence & Data Routing

Read-Replica Routing

`ReadWriteRoutingDataSource` automatically routes
`@Transactional(readOnly = true)` calls to the replica
connection pool via `LazyConnectionDataSourceProxy`.

Liquibase Migrations

Strict version control with one changeset per XML file.
Liquibase directly binds to the primary database URL to
execute migrations, completely bypassing read-replica proxy
routing.

System Metrics & Footprint

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Active REST Endpoints

Engineered for Scale & Maintainability

The platform encompasses **20 Controllers**, **24 Service Implementations**, **20 Spring Data Repositories**, and **43 JPA Entity types**.

Configuration settings are strongly typed via `@ConfigurationProperties` classes, covering dynamic pricing thresholds, seat hold TTLs, and refund policies.

Questions & Discussion

Thank you! For endpoint-level details, refer to the functional specification.

`GET /v3/api-docs | swagger-ui.html`

Image Sources



<https://atscada.com/wp-content/uploads/2014/10/ATSCADA-Software-Architecture-e1776224330803.jpg>

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